

The Bayesian Algorithm

Lancaster, Ch. 1 — core deck

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An Introduction to Modern Bayesian Econometrics (2004) — reading group draft

The whole book, in five steps

ALGORITHM 1.1 — Bayes

1. Write down a **likelihood**.
2. Write down a **prior**.
3. Put the data in.
4. Turn the crank — Bayes' theorem.
5. **Criticize your model.**

Steps 1–4 are Chapter 1. Step 5 is Chapter 2.

Everything after that — regression, MCMC, panels, time series — is these same five steps on harder models.

Two sessions

Session 1 — the machine

Appendix 1 · Bayes' theorem · the algorithm · the likelihood · exchangeability · prior · posterior

Session 2 — what you do with it

Monte Carlo and ABC · prediction · decisions and Wald · value at risk

Detail, quotes and derivations live in the extended deck.

I. What are we even doing?

The confrontation of an economic model with evidence.

It asks two questions:

1. **Is the model consistent with the evidence?** → Chapter 2
2. **If so, which structures are probable?** → Chapter 1

In practice we ask them backwards: assume it fits, then find the probable structures.

Econometrics is not statistics

Statistics

Reduce a pile of numbers to something a mind can hold.

Data first.

Econometrics

Explain the behavior of **agents** in markets.

Data second.

That is why our models have equilibrium, optimization, and agents who know things we don't — and why our parameters mean something.

Ask the room

An elasticity of substitution exists only inside a theory. **In what sense can such a thing be *true*?**

II. Appendix 1 — the conversion manual

The frequentist picture

Everyone here was trained this way. Appendix 1 is not an argument — **it is a map of what changes.**

- There is a **population**. Your data are a **sample** from it.
- There is a **true** θ_0 .
- You pick an **estimator** — a formula.
- You judge that formula by how it behaves **over samples you never saw**.

Bias, variance, MSE, power, consistency: **every one is an average over imaginary data.**

The Bayesian picture

- You have **beliefs** about θ , and data in hand.
- Bayes' theorem tells you how the data **change** the beliefs.
- You report the resulting distribution.

The one thing

Frequentist inference conditions on a θ you never learn.

Bayesian inference conditions on the data you actually have.

What actually changes

Four of Lancaster's eight

1. Bayes **demands more**: a full likelihood, not just a moment condition.
2. There is only **one estimator** — the posterior. Model and data imply it.
3. Bias, MSE, efficiency are **irrelevant** — including Gauss–Markov.
4. Consistency asks what happens **with more data than you have**.

The bus example

200 Bay Area commuters. You toss a coin for each: bus or car.

You end up with **97 bus, 103 car**.

Now the frequentist must decide what to imagine repeating:

- samples of size 97 and 103?
- or 200 coin tosses, with the counts moving around?

The standard errors differ. The Bayesian conditions on 97 and 103 and never notices.

Ask the room

Ask the room

Your standard errors changed because of a coin toss that has nothing to do with commuting.

Cox & Hinkley say: condition on what you actually saw.

Once you start conditioning, how far can you go before you have become a Bayesian?

Sometimes the numbers agree

n normal draws, known variance. Both camps report the same interval.

Frequentist

“95% of intervals built this way would contain μ .”

A claim about the **procedure**.

Bayesian

“The probability that μ is in here is 0.95.”

A claim about μ .

Same numbers. Only one of them is the sentence you actually wanted to say.

III. The algorithm

Bayes' theorem

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta)$$

Posterior $\propto$ **likelihood** $\times$ **prior**.

Nobody disputes the theorem. The disputes are about two other things:

- what probability **means**
- what it may be **applied to**

Probability means degree of belief

The larger $P(A)$, the more strongly you believe A .

That is what lets an analysis end with:

“In the light of the evidence, theory A is very unlikely and theory B is quite probable.”

In traditional econometrics that sentence is illegal. Theories are true or false, not probable.

What may have a probability?

Frequentists: only things embedded in a repeatable sequence.

Lancaster: anything you have beliefs about.

- the US was in recession in 1983
- the moon is made of green cheese
- the economy is in general equilibrium

There is no sequence of worlds in some of which 1983 was a recession.

Evidence moving a belief

Two structures. You think each is 50/50.

The data are **unlikely** under structure 1 (0.1), **likely** under structure 2 (0.6).

prior odds 1 : 1 $\longrightarrow$ **posterior odds** 6 : 1

The evidence entered as a **ratio**, 0.6/0.1. Nothing else about the data mattered.

Two ways evidence fails to move you

You were dogmatic

If you gave a theory probability zero, no data can ever raise it.

"If you gave no credence at all to a theory you will never learn that it is right."

The data can't tell them apart

If both structures make the data equally probable, the data say nothing.

Make the second one true for *every possible dataset* and you have **non-identification**.

Parameters and data

The only distinction that matters

A **parameter** is unknown to you before the data *and after*.

Data are unknown before and **known after**.

That's it. So a parameter can be a constant, a whole theory, or an unknown function.

And “random” vs. “fixed” stops meaning anything: **everything unknown gets a distribution.**

The algorithm again

ALGORITHM 1.1

1. likelihood 2. prior 3. data 4. Bayes' theorem 5. criticize

Steps 1–4 are mechanical once you commit.

Step 5 is the only step where the world gets to talk back.

Which is why Chapter 2 is not an appendix to Chapter 1. It is the other half.

IV. The likelihood — the information transfer function

One object, two names

$$p(y \mid \theta) \quad \text{vs.} \quad \ell(\theta; y)$$

- Before the data: a **distribution over data**, given θ .
- After the data: a **function of θ** , with your data baked in.

Note the semicolon. It is **not** a probability distribution for θ .

The one thing

This is the only door the data come through. Prior in, posterior out — the likelihood is what does the transferring.

Choosing a likelihood commits you

It must:

1. **express your economic model**
2. let you say which parameter values are probable
3. **let the data prove you wrong**

Ask the room

Write down the likelihood for your current project.

Now write down a dataset that would embarrass it.

If you can't, what have you actually assumed?

The likelihood principle

Two researchers. Same coin. Same prior.

A decides to do 20 trials.

Gets 7 successes.

B decides to stop at 7 successes.

The 7th lands on trial 20.

Different experiments, different distributions — **same likelihood, up to a constant.**

So: **same posterior, same inference.**

Which means the stopping rule is irrelevant

It does not matter whether you planned 20 trials, planned to stop at 7 successes, or just got bored and quit.

The frequentist answer **depends on what was in the researcher's head.**

Ask the room

Would you throw away unrepeatable data because you couldn't find out what someone intended when they started?

This is what converted Savage

Savage spent the 1950s trying to rebuild statistics on Wald's foundations.

The project failed — publicly, in his own words.

He said his conversion to Bayesianism came from **discovering the likelihood principle.**

The idea that shocks a first-year class is the one that turned the most careful frequentist mind of the era into “*we radical Bayesians.*”

V. Exchangeability

The problem with “iid”

Every model starts with “ $Y_1, \dots, Y_n$ are independent and identically distributed.”

But that sentence implies a **machine out there** producing draws — probability with an objective existence.

If probability is your degree of belief, what is that sentence even saying?

Definition 1.4

Your beliefs are **exchangeable** if reordering the observations doesn't change your joint distribution.

- iid $\Rightarrow$ exchangeable. **Not the reverse.**
- It is a **judgement**, and it is yours.

Notice the grammar: the coin isn't exchangeable. **Your beliefs are.**

If your beliefs about a binary sequence are exchangeable, then

$$p(y_1, \dots, y_n) = \int \theta^s (1 - \theta)^{n-s} dF(\theta)$$

Look at what's on the right: **a likelihood, integrated against a prior.**

The one thing

You never needed to assume iid. Judge your beliefs symmetric, and a likelihood *and* a *prior* fall out whether you wanted them or not.

Ask the room

Ask the room

Are daily returns on SPY exchangeable in your beliefs?

If a 2008 observation and a 2017 observation were swapped, would anything change?

If yes — you have just signed up to model the dependence.

VI. The prior

It's yours, and you don't have to believe it

You should try other priors on purpose: *“if I had believed this, what would I now believe?”*

That's **sensitivity analysis**.

You may even look at the data first. Bayes' theorem never told you what to believe — only how beliefs **change**.

Two rules of thumb

Encompass your audience

A prior nobody shares makes your paper uninteresting: *"So what, I don't believe A."*

Use one that isn't sharply inconsistent with any reasonable belief.

Never assign zero

Zero times anything is zero. You can never learn that the truth was there.

Conjugacy is a convenience

If the prior and posterior are in the same family, you never do an integral.

Bernoulli likelihood $\times$ beta prior $\rightarrow$ beta posterior:

```
using Distributions
prior      = Beta(1, 1)           # uniform: 0 prior successes, 0 failures
n, s      = 20, 7                # 20 trials, 7 successes
posterior = Beta(1 + s, 1 + n - s) # just add them up
mean(posterior), std(posterior)   # 0.3636, 0.1003
```

Updating is counting. The prior is a pile of imaginary earlier trials.

Conjugacy mattered when we did this by hand. Now it's a nicety.

Where the prior stops mattering

The likelihood is essentially **zero almost everywhere**.

So your prior gets multiplied by zero almost everywhere. What you believed out there is irrelevant.

The one thing

Least squares and maximum likelihood are flat-prior Bayes.

So what exactly were you objecting to?

VII. The posterior

1. **Draw it.** For a scalar — or a scalar function of a vector — nothing beats the picture.
2. **Report moments.** Posterior mean and SD. Looks like your usual table.
3. **Report an HPD interval.** The shortest interval holding 95% of your belief.
4. **Report marginals.** Integrate the rest out.

Only #4 is hard. Part VIII is about that.

The sentence you are allowed to say

Confidence interval

“If I repeated this procedure many times, 95% of the intervals I’d build would cover the true value.”

HPD interval

“The probability that ρ is between 0.94 and 1.08 is 0.95.”

It’s as simple as that.

Every student who has ever misinterpreted a confidence interval was reaching for the right-hand sentence.

Data wins in the end

The log posterior is the log likelihood plus the log prior.

Data accumulate; the likelihood term grows. **The prior term just sits there.**

Three very different priors, 5 observations: still visibly different.

Same three priors, 20 observations: nearly identical.

Two honest people are dragged into agreement by evidence — **as long as neither was dogmatic, and both agree on the likelihood.**

Sampling the posterior

Stop trying to do the integrals. **Draw from the posterior instead.**

One row per draw, one column per parameter:

- the whole matrix is the **joint** posterior
- one column is a **marginal** — to study θ_1 , *ignore the other column*
- apply any function **row-wise** and you have that function's posterior

The one thing

If you can sample a distribution, you can know it as precisely as you like. This is the whole modern shift — computer power in place of hard mathematics.

Which lets you answer the actual question

Want the marginal effect, the elasticity, the probability an effect is positive?

Compute it on **every draw**. You now have its posterior. Read off whatever you want.

Ask the room

"The posterior probability that this effect is positive is 0.94."

Write down the frequentist object that answers the same question.

Take your time.

VIII. When there's no conjugate prior

Two integrals stand in the way

$$p(y) = \int p(y \mid \theta) p(\theta) d\theta \qquad p(\theta_j \mid y) = \int p(\theta \mid y) d\theta_{-j}$$

Conjugacy made the first one disappear — you *recognized* the answer instead of computing it.

The second is the marginalization. In general: hard.

Two escapes: approximate the posterior, or **sample** it. Sampling wins — with an approximation you can never check the error without doing the exact calculation you were avoiding.

Monte Carlo is integration by sampling

Every posterior summary is an average of *something* over the posterior:

You want	Average this over the draws
posterior mean	θ
$P(\theta > 0 \mid y)$	1 if $\theta > 0$, else 0
a marginal effect	the effect, computed per draw
value at risk	a quantile — just sort them

You never needed a formula for the posterior. You needed draws from it.

ABC: when you can simulate but not evaluate

You can run your model **forward** — give it a θ , it spits out plausible data.

You cannot write down $p(y \mid \theta)$.

This is normal life for:

- structural and agent-based models
- numerically-solved equilibrium models
- anything with a simulator and no formula

ABC is the algorithm, run forward

Rejection ABC

1. Draw θ from the **prior**.
2. **Simulate** a dataset from it.
3. Keep θ if the simulated data look like the real data.

```
draws = Float64[]  
for m in 1:400_000  
    ts = rand(Uniform(0, 1))      # 1. from the prior  
    ys = rand(Binomial(20, ts))   # 2. SIMULATE (no pdf evaluated)  
    ys == 7 && push!(draws, ts)    # 3. keep it if it matches  
end  
mean(draws), std(draws)          # 0.3636, 0.1003  <- the exact answer
```

Two things that just happened

It was exact, not approximate

The number of successes is **sufficient**, and we demanded an exact match. So we conditioned on everything the data contained — and got the conjugate answer without ever writing a likelihood.

The acceptance rate is $p(y)$

$1/21 = 4.76\%$ of draws survived. That fraction *is* the marginal likelihood — the denominator we spent all of Chapter 1 throwing away.

And Chapter 2 uses it to check models and choose between them.

What ABC costs you

- **Sufficient statistics are rare.** Usually you pick summaries — so you're answering a coarser question than you asked.
- **Loosening the match widens the posterior.** Report your tolerance or you're reporting an estimate with no standard error.
- **It doesn't scale** in the number of summaries.

Ask the room

ABC never evaluates a likelihood — but you chose the summary statistic in advance.

Is that different from choosing a test statistic?

IX. Prediction

It was there the whole time

The denominator we kept dropping:

$$p(y) = \int p(y \mid \theta) p(\theta) d\theta$$

Lancaster gives it two names: the **marginal** distribution of the data, and the **predictive** distribution of the data.

The one thing

Writing down a likelihood and a prior *is* predicting your data.

Model criticism is just comparing that prediction to what showed up.

Two predictive distributions

Prior predictive

What your data **should have looked like**, before you saw them.

Basis of formal model checks.

Posterior predictive

What **another draw** from your model should look like, now that you've seen the data.

A forecast, or a replication of your own dataset.

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta) p(\theta | y) d\theta$$

How to predict — Algorithm 2.5

Two steps, repeated

1. Draw a θ from the posterior.
2. Draw a $\tilde{y}$ from the model at that θ .

One θ per prediction.

Your uncertainty about the parameter and the world's randomness get compounded — not stacked side by side, not reported separately.

The plug-in fallacy

Standard practice: estimate $\hat{\theta}$, then predict as if it were correct.

Lancaster: this **has no theoretical justification** except as an approximation — and only when the posterior is tightly concentrated.

Plug-in prediction is always too confident. It reports the world's variability and quietly drops yours.

In the AR(1) example, the predictive precision is *a/ways* less than the model's — and it gets worse when you forecast from an extreme starting point.

Prediction is the primitive

de Finetti, one more time:

The thing you really have beliefs about is **the observables**.

θ is a device for representing those beliefs compactly.

Ask the room

Which would you rather defend in a seminar — your posterior for β , or your prediction for next quarter?

Which one can actually be checked?

So why do our tables report the other one?

X. Decisions — and Wald

Lancaster declines the decision frame

Many writers treat inference as **decision making**. Economists like that: their agents optimize.

Lancaster says no. The econometrician's job is **reporting**:

Draw the marginal of the object of interest, and leave it to others — policy makers — to decide.

But loss functions are already there

Given a **loss function**, the Bayes decision minimizes **expected loss over the posterior**.

- squared error loss $\rightarrow$ report the posterior **mean**
- absolute error $\rightarrow$ the **median**
- 0–1 loss $\rightarrow$ the **mode**

The one thing

The "estimate" you report is a statement about your loss function — not about your beliefs.

An example with teeth

Data uniform on 0 to θ . Under squared error loss, the Bayes decision is:

take the largest observation and nudge it up.

Maximum likelihood reports the largest observation itself — which is **too low for essentially every dataset that could ever occur.**

The posterior didn't change between those two answers. Only the loss function did.

Wald: statistics *is* decision theory

Marschak, 1946: von Neumann says *to be an economic man you must be a statistical man*. Wald was arguing the reverse.

Wald's statistical decision problem: actions, states of the world, a **loss function**, and an experiment that costs something.

A statistical answer should tell you **what to do**, not what to say. He called it *inductive behavior*.

And since Nature's choice is unknown, act as if Nature were trying to hurt you — **minimax**, statistics as a game.

The complete class theorem (Wald, 1947)

For any non-Bayesian decision rule there is a **Bayesian** rule at least as good.

The Bayesian rules are *complete*.

Wald proved it — and then used priors “merely as mathematical tools,” denying they meant anything.

The most rigorous frequentist decision theorist of the century proved you never need to be non-Bayesian, and declined to be a Bayesian.

Discussion: econometrics vs. decision theory

For the room

1. Giocoli's paradox: we model our **agents** as Bayesians and do our own estimation as frequentists. Inconsistency, or division of labor?
2. Lancaster's answer: emphasize both perspectives "**not because econometricians take decisions but because agents do.**" Resolution or truce?
3. "Just report the marginal" — but a report has a length, a format, and things left out. Is that really free of a loss function?
4. If a central bank or a risk committee will act on your number, are you still only reporting?

XI. Bridge to Chapter 2 — value at risk

Where Chapter 2 goes

- 2.1–2.2 How to check a model. QQ plots, thick tails, mixtures — *i.e.* returns.
- 2.3 Which beliefs the data can never check.
- 2.4 Formal checks, via the **prior predictive**.
- 2.5 **Posterior prediction**.
- 2.6 Posterior odds, Bayes factors, model averaging.
- 2.7 Enlarging the model — constant variance → **ARCH**.

Chapter 2 opens: “**If you don’t criticize your results you can be sure that others will.**”

VaR and ES are just facts about a distribution

- **Value at risk:** a **quantile** of the loss distribution.
- **Expected shortfall:** the **average loss beyond** that quantile.

Neither is hard. The entire question is:

The one thing

Which distribution?

The model at your estimate — or the posterior predictive?

Two sources of uncertainty

$$p(\tilde{r} | y) = \int \underbrace{p(\tilde{r} | \theta)}_{\text{tomorrow's shock}} \underbrace{p(\theta | y)}_{\text{your ignorance}} d\theta$$

Plug-in keeps the first and drops the second.

Do it properly for normal returns and the predictive distribution comes out **Student- t** , not normal.

The one thing

Some of the fat tail in your risk model isn't the market. It's you, not knowing θ .

How much does plug-in understate the risk?

Tail multipliers, in units of s :

n	level	VaR plug-in	VaR	ES plug-in	ES predictive
			predictive		
25	1%	2.33	2.54	2.67	2.98
60	1%	2.33	2.41	2.67	2.79
250	1%	2.33	2.35	2.67	2.69

Biggest correction exactly where risk management is hardest: **short samples, deep tails.**

And ES is corrected more than VaR.

Predictive VaR in three lines

```
# draw parameters from the posterior, then returns from the model
sigma2 = nu * s2 ./ rand(Chisq(nu), M)           # step 1
mu      = rbar .+ sqrt.(sigma2 ./ n) .* randn(M)
rtilde  = mu .+ sqrt.(sigma2) .* randn(M)        # step 2

q       = quantile(rtilde, 0.01)
VaR     = -q
ES      = -mean(rtilde[rtilde .<= q])
```

Same two steps as Algorithm 2.5 — **and this works when the model is a GARCH, a jump diffusion, or a simulator you can only run forward.**

VaR itself has a posterior

Predictive VaR — one number.

Your uncertainty is already averaged in.

This is what you hold capital against.

Posterior of VaR — a whole distribution.

VaR is a function of θ , so compute it on every draw.

60 days of returns, 1% level: point estimate **0.0247**, but the posterior says

[0.0205, 0.0304]

Your VaR is known to about $\pm 20\%$. The report shows three decimals and no interval.

Ask the room

Ask the room

1. Which number does a capital requirement need — and which does a risk *committee* need?
2. Backtesting counts exceedances over many days. That is a repeated-sampling check on a Bayesian number. Incoherent, or exactly right?
3. Expected shortfall is an **expected loss**. That is Wald's risk function wearing a Basel hat. **Did the regulators become Bayesians, or Waldians?**

Are returns exchangeable? **No** — 2008 and 2017 are not swappable.

So you model the dependence: that's ARCH, GARCH, stochastic volatility. Chapter 2's "enlarging the model."

Then place your exchangeability judgement one level down — on the **standardized** returns.

That judgement is checkable. And it is where the tail risk lives.

Close

ALGORITHM 1.1

1. likelihood 2. prior 3. data 4. Bayes' theorem 5. criticize

- **Likelihood** — the only door the data come through.
- **Exchangeability** — judge your beliefs symmetric, and you already have a prior.
- **Prior** — yours, revisable, never dogmatic.
- **Posterior** — don't integrate it. **Sample it.**
- **Prediction** — the observables are what you can actually check.
- $p(y)$, the thing we kept dropping, is the predictive distribution — and it's where Chapter 2, model choice, and every risk number live.

Take these with you

1. Rewrite one table from a paper you admire as a Bayesian would report it.
2. Write down a dataset that would embarrass your own model.
3. Simulate the two stopping rules. Confirm the posteriors agree and the p-values don't.
4. Decide the level at which you'd call your data exchangeable, and defend it.
5. Compute plug-in VaR, predictive VaR, and the posterior of VaR — then argue for the number you'd hand a risk committee.

Sources: Lancaster (2004) Ch. 1, Ch. 2, Appendix 1 · Giocoli (2013), *From Wald to Savage* · full detail in the extended deck.